Report Date: April 02, 2020

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Report Number: 147158

Town of Ladysmith
PO Box 220, 410 Esplanade
Ladysmith, BC
V9G 1A2

Dear Town of Ladysmith,

Re: Non-compliance Advisory Letter, Permit 120, 245 A Oyster Cove Road, Ladysmith, BC Municipal Wastewater Treatment Plant

On March 31, 2020, Ministry of Environment and Climate Change Strategy, Environmental Protection Division staff conducted an inspection of the Town of Ladysmith's wastewater treatment facility located at 245 A Oyster Cove Road, Ladysmith, BC with authorization number 120 under the *Environmental Management Act*.

This Advisory, the alleged violations and the circumstances to which it refers will form part of the compliance history of Town of Ladysmith, and will be taken into account in the event of future non-compliance.

Please note that this authorization is considered to be out of compliance until such a time as it can be confirmed to meet the authorization requirements.

Inspection Details:

Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.1: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.1 The maximum rate at which effluent may be discharged through the sewage treatment plant is 6 100 m3/d; the annual average rate of discharge is 3 050 m3/d. Flow in excess of treatment plant capacity is to be discharged to the bypass outfall. The maximum rate at which effluent may be bypassed is 8 500 m3/d; the total annual volume which may be discharged to the bypass outfall is 228 000 m3.

Ministry of Environment and Climate Change Strategy

Compliance
Environmental Protection Division

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Details/Findings:	On March 31, 2020, Ministry of Environment and Climate Change Strategy (Ministry) Officer Connor Fraleigh (Officer Fraleigh) conducted an office review inspection of the Town of Ladysmith wastewater treatment facility (Facility) located in Ladysmith, BC to verify compliance with Permit 120 (Permit). The Permit authorizes the discharge of effluent to Ladysmith Harbour from a municipal sewage system. The Permit was first issued on March 12, 1965, and last amended on January 10, 1994. The inspection period for this report is from January 1, 2019 to December 31, 2019 and included a review of the following documents: - 2019 Quarterly Water Quality Data Reports (2019 Q1, Q2, Q3, Q4 WQ DATA) - 2019 Summary of Bypasses at the Wastewater Treatment Plant report was submitted to the Ministry on January 10, 2020 (2019 Bypass Report) A review of the 2019 WQ Data confirmed that the Town of Ladysmith exceeded the maximum allowable rate of discharge on the following dates: January 3, 2019 - 11,240 m3/d January 4, 2019 - 9,424 m3/d Flow rates for the remainder of the inspection period were within the allowable maximum daily discharge rate.
Compliance:	Out
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.1: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.1 The maximum rate at which effluent may be discharged through the sewage treatment plant is 6 100 m3/d; the annual average rate of discharge is 3 050 m3/d. Flow in excess of treatment plant capacity is to be discharged to the bypass outfall. The maximum rate at which effluent may be bypassed is 8 500 m3/d; the total annual volume which may be discharged to the bypass outfall is 228 000 m3.
Details/Findings:	A review of the 2019 WQ Data confirmed that the annual average rate of discharge was in compliance with the permitted maximum. The annual average rate of discharge was calculated by Officer Fraleigh as 2,490 m3/d.
Compliance:	In

Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.2: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.2 The characteristics of the discharge shall not exceed: Sewage Treatment Plant Discharge: 5-day Biochemical Oxygen Demand: 130 mg/L. Total Suspended Solids: 130 mg/L. Overflow Bypass Discharge: Screened combined stormwater and sewage effluent.
Details/Findings:	A review of the 2019 WQ Data confirmed that there were no exceedances for the characteristics of the discharge throughout the inspection period. All reported values during the inspection period for 5-day Biochemical Oxygen Demand (BOD5) and Total Suspended Solids (TSS) were well below permitted limits. The maximum values reported during 2019 were BOD5 = 14 mg/L (June) and TSS = 14 mg/L (December).
Compliance:	In
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.3: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.3 The works authorized are influent screening, a Spiragester primary type treatment plant, chlorination and dechlorination facilities, autothermophilic aerobic sludge processing facilities, foul air scrubber, sludge dewatering facilities, overflow bypass, bypass screening and solids removal facilities, 300 mm and 710 mm diameter parallel outfalls extending 850 m from mean low water to a depth of 17.9 m below mean low water, and related appurtenances approximately located as shown on the attached Site Plan A.
Details/Findings:	The authorized works were not evaluated as part of this office review inspection. Therefore, compliance with this requirement was not applicable.
Compliance:	Not Applicable

Requirement Description:	**2. GENERAL REQUIREMENTS, 2.1 Maintenance of Works and Emergency Procedures** 2.1: The Permittee shall inspect the pollution control works regularly and maintain them in good working order. In the event of an emergency or condition beyond the control of the Permittee which prevents continuing operation of the approved method of pollution control, the Permittee shall immediately notify the Regional Waste Manager and take appropriate remedial action.
Details/Findings:	The requirement to maintain the authorized works was not assessed as part of this office review inspection. Therefore, compliance with this requirement was not applicable.
Compliance:	Not Applicable
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.2 Bypasses** 2.2: The discharge of effluent which has bypassed the designated treatment works is prohibited unless the consent of the Regional Waste Manager is obtained and confirmed in writing.
Details/Findings:	Page one of the 2019 Bypass Report states that there were no bypass events to report in 2019. Therefore, compliance with this requirement is not applicable throughout the inspection period.
Compliance:	Not Applicable
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.5 Outfall Inspection** 2.5: The Permittee shall conduct a dye test on the outfall lines (or inspect by another method approved by the Regional Waste Manager) every five years or as may otherwise be required by the Regional Waste Manager.
Details/Findings:	The last dye test conducted on the Town of Ladysmith's wastewater treatment plant outfalls was completed on February 21, 2018. Therefore, compliance with this requirement was not applicable throughout the inspection period.
Compliance:	Not Applicable

Requirement Description:	**2. GENERAL REQUIREMENTS, 2.6 Chlorination** 2.6: Maintain a chlorine residual (at the point of discharge or prior to dechlorination) between 0.1 and 1.0 mg/L at all times.
Details/Findings:	A review of the 2019 WQ Data confirmed that the chlorine residual values (at the point of discharge or prior to de-chlorination) were maintained between 0.1 and 1.0 mg/L throughout the inspection period.
Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.1 Sampling and Analysis** 3.1.1: The Permittee shall install a suitable sampling facility and obtain a composite sample of the effluent from the treatment plant effluent once every month. The sample is to consist of four grab samples taken over a two-hour period at maximum flow and mixed to form a single sample for subsequent analysis. Proper care should be taken in sampling, storing and transporting the sample to adequately control temperature and avoid contamination, breakage, etc. Obtain analyses of the sample for the following: 5-day Biochemical Oxygen Demand; Total Suspended Solids.
Details/Findings:	Following a review of the 2019 WQ Data, it was confirmed that sampling and analyses were performed for 5-day Biochemical Oxygen Demand and Total Suspended Solids as required. Fecal coliforms samples were also collected an analyzed on a monthly basis. Sample analysis was performed by Maxxam Analytics and ALS Environmental of Burnaby, BC.
Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.2 Flow Measurement** 3.1.2: Provide and maintain suitable flow measuring devices and record once per day the effluent volume discharged over a 24-hour period from the sewage treatment plant and the overflow bypass.
Details/Findings:	Following a review of the 2019 WQ Data, it was confirmed that the Town of Ladysmith recorded and submitted effluent volumes discharged over a 24-hour period from the sewage treatment plant throughout the inspection period.

Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.4 Reporting** 3.4: Maintain data of analyses and flow measurements for inspection and every three months submit the data, suitably tabulated, to the Regional Waste Manager for the previous quarter. The first report is to be submitted by January 31, 1994. Based on the results of the monitoring program, the Permittee monitoring requirements may be extended or altered by the Regional Waste Manager.
Details/Findings:	Officer Fraleigh confirmed that all 2019 data of analyses and flow measurements were submitted to the Ministry as required by this section.
Compliance:	In

The scope of this inspection was a monitoring data review only, completed for reporting purposes related to the annual Conditional Management Plan (CMP) which is to provide enhanced management of the molluscan shellfish harvest in waters adjacent to the Town of Ladysmith wastewater treatment facility.

Compliance History:
2019-01-28 - IR115167 - Advisory - discharge rate exceedances, unauthorized bypass,
2018-01-23 - IR76276 - Advisory - discharge rate exceedances, characteristic (BOD5) exceedances, chlorine residuals outside of required range and a lapse of outfall inspection frequency.
2017-03-02 - IR49542 - Advisory - EOCP operator not qualified

The Ministry of Environment Compliance and Enforcement Policy and Procedure (C&E Policy) prescribes common requirements and procedures for all Ministry staff to ensure consistent and risk-based assessment and response to non-compliance. Using the Non-Compliance Decision Matrix, the compliance determination for this inspection has been assess as a **Level 1, Category B, Advisory**.

More information about Environmental Compliance, the Non-Compliance Decision Matrix, and reporting and data submission requirements can be found at the links below.

General compliance information:
www.gov.bc.ca/environmentalcompliance

Non-Compliance Decision Matrix information:
https://www2.gov.bc.ca/gov/content/environment/natural-resource-stewardship/natural-resource-law-enforcement/environmental-compliance/how-compliance-is-assessed

Reporting and data submission requirements (to be sent to EnvAuthorizationsReporting@gov.bc.ca):
https://www2.gov.bc.ca/gov/content/environment/waste-management/waste-discharge-authorization/data-and-report-submissions/routine-environmental-reporting-submission-mailbox

Please be advised that this inspection report may be published on the provincial government website within 7 days.

If you have any questions about this letter, please contact the undersigned.

Yours truly,

Connor Fraleigh
Environmental Protection Officer

cc: Mike Brown
Curtis Baker

Ministry of Environment and Climate Change Strategy
Compliance
Environmental Protection Division
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DISCLAIMER:
Please note that sections of the permit, regulation or code of practice referenced in this inspection record are for guidance and are not the official version. Please refer to the original permit, regulation or code of practice.

To see the most up to date version of the regulations and codes of practices please visit http://www.bclaws.ca

If you require a copy of the original permit, please contact the inspector noted on this inspection record.

It is also important to note that this inspection record does not necessarily reflect each requirement or condition of the authorization therefore compliance is noted only for the requirements or conditions listed in the inspection record.